

JONATHAN DCRUZ

Systems Engineer | Technical Product Manager | Systems Integration & Cross-Functional Development

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PROFESSIONAL SUMMARY

Systems-minded engineer who translates complex hardware, networking, and IoT architectures into clean, user-centered software — treating integration, not isolated components, as the core engineering problem to solve. Operates with a big-picture architectural mindset, orchestrating developers, AI-assisted tools, and interdependent system components so that fragmented technical efforts converge into unified, working products. Functions as the connective layer on technical teams rather than a generalist spread thin — equally fluent in hardware constraints, network protocols, and product vision, with the ability to drive a system from concept through cross-functional execution to a working end-to-end result.

CORE COMPETENCIES

- Systems Engineering & Integration
- Technical Project Coordination
- UI/UX Design & Product Vision
- Hardware & IoT Networking
- Prompt Engineering & Rapid Prototyping
- Avionics, Aeronautics & Space Tech

KEY PROJECTS & ACHIEVEMENTS

Vehicle Tailgating Prevention System — IoT/Hardware Systems Project

Winner, GEMS Arab Innovation Centre for Education (AICE) Accelerator Programme — 2018 GEMS Innovation Awards

- Engineered an IoT-based vehicle tailgating detection and prevention system, using networked routers to link peripheral hardware devices into a coordinated sensing and response architecture.
- Applied network protocol design to enable real-time communication between distributed hardware components, allowing the system to detect and flag tailgating incidents as they occurred.
- Selected as a winning project in the AICE Accelerator Programme category, one of four competition tracks at the 2018 GEMS Innovation Awards, recognized among 200+ student innovators across the GEMS Education network. (Coverage: Gulf News, COTTM)

Reading Disability Evaluation System — Applied ML & Networking

Technical Coordination & Systems Development

- Coordinated end-to-end development of a locally-hosted, hardware-optimized evaluation system, balancing on-device computational constraints with real-world usability requirements.
- Integrated Machine Learning algorithms to drive the system's core evaluation logic, alongside WebRTC networking protocols to support real-time, low-latency data transmission.
- Directed collaboration between algorithmic and infrastructure work streams, ensuring the ML pipeline and networking layer functioned as a single coherent system rather than disjointed parts.

Hyper-Specific App Development — AI-Assisted Product Delivery

Rapid Prototyping & Prompt Engineering

- Applied advanced prompt engineering techniques to independently architect, build, and deploy custom applications tailored to narrow, specific use cases.
- Used AI-assisted development tooling to compress the concept-to-deployment cycle, enabling functional products to be shipped without a traditional multi-engineer build team.
- Demonstrated the ability to single-handedly own a product from requirements through deployment — a capability that underpins a broader systems-integration approach to product work.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering

Mar Baselios College of Engineering and Technology (MBCET) — Currently Pursuing | Expected Graduation: 2027

GitHub Portfolio: github.com/F14xtomcat